A Fully Automatic Hand-Eye Calibration System

Morris Antonello, Andrea Gobbi, Stefano Michieletto, Stefano Ghidoni and Emanuele Menegatti

***Abstract*— To retrieve the 3D coordinates of an object in the robot workspace is a fundamental capability for industrial and service applications. This can be achieved by means of a camera mounted on the robot end-effector only if the hand-eye transformation is known. The standard calibration process requires to view a calibration pattern, e.g. a checkerboard, from several different perspectives. This work extends the standard approach performing calibration pattern localization and hand-eye calibration in a fully automatic way. A two phase procedure has been developed and tested in both simulated and real scenarios, demonstrating that the automatic calibration reaches the same performance level of a standard procedure, while avoiding any human intervention. As a final contribution, the source code for an automatic and robust calibration is released.**

I. INTRODUCTION

The hand-eye calibration problem consists in estimating the rotation and translation of the end effector or gripper of a robot, i.e. the hand, with respect to the camera mounted on it, i.e. the eye. As reviewed in [1], a number of typical robot tasks like visual servoing [2], grasping and stereo vision requires the knowledge of this transformation or, at least, could highly benefit from it. This way, the robotic arm can move the camera with respect to a fixed reference system, e.g. the robot base or the workcell world reference. Unfortunately, in most applications the hand-eye transformation needs to be calculated several times during the life cycle of a system, thus requiring many manual interventions.

A well-known approach was proposed by Tsai et al. [1] in 1988. Such approach requires the user to acquire a large set of images (more than 50) framing a calibration pattern (usually a checkerboard) from different views and the corresponding set of robot poses. Its implementation is already available in the Robot Operating System (ROS) [3], a framework widely used in robotics, and in the Visual Servoing Platform (ViSP) [4], a library for developing visual servoing systems. This task can be partially automatized by fixing the pattern position inside the robotic work-cell and saving a pre-determined robot motion. Even if this can be easily done with recent robotic arms, there are some drawbacks: it might not be possible to fix the pattern position, and this solution would not work after any change in the work-cell or robot-camera configuration. An automatic approach automatizing the robot motion has been proposed by Tsai et al. [5] but, given that it requires the pattern calibration, i.e. the knowledge of the pattern position in the world, it shares the same drawbacks. Indeed, the pattern localization can be cumbersome. It is usually achieved by means of two standard procedures described in the robot vendor manuals: the calibration of a tip mounted on the end effector and the calibration of the work object, in this case the calibration pattern. The tip allows to accurately touch three points on the calibration pattern so as to define a Cartesian reference system on it.

In this paper, such standard approach [1], [5] is extended so as to perform both the calibration pattern localization and hand-eye calibration in a fully automatic way. This is a chicken-and-egg problem: to localize the pattern by means of the camera, the hand-eye calibration is needed; however, in order to automatize the hand-eye calibration, the pattern localization is also needed. Here, this is overcome by means of two phases, in each of which an hand-eye calibration is performed as shown in the overview in Figure 1. In particular, our main contributions are:

- A fully automatic hand-eye calibration procedure, which does not require the knowledge of the calibration pattern 3D location;
- An open-source module[1] based on the framework ROS and the libraries ViSP and MoveIT! [6], which can be easily exploited on other robot platforms.

These results could speed up the research of the wide community working with robotic manipulators on complex and precise manufacturing tasks. In the challenge EuRoC[2][7], the hand-eye transformation was necessary for visual servoing in tasks like car door assembly [8] and coil winding [9], [10]. In the challenge MBZIRC[3], many mobile manipulators were equipped with a camera on the robotic arm. The hand-eye transformation was necessary for grasping a wrench and rotating a valve stem. Furthermore, from our experience, the hand-eye transformation turns out to be useful also in other specific scenarios. In the context of the EU project FibreMap[4][11], it allowed to inspect a carbon fiber preform in order to analyze its surface with a high accuracy and map carbon fibres on its model [12]. In ThermoBot[5], it allowed to subtract the background from thermographic images, thus improving the speed and reducing false positives when detecting defects in carbon fiber parts [13]. Different examples of robot hand-eye configurations are reported in Figure 2. In our real experiments, the configuration in Figure 2(a) has been adopted.

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[1] `http://robotics.dei.unipd.it/129-auto-hand-eye`
[2] `http://www.euroc-project.eu/`
[3] `http://www.mbzirc.com/`
[4] `http://fibremap.eu/`
[5] `http://thermobot.eu/`

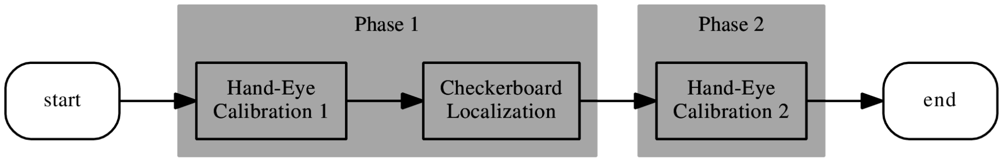
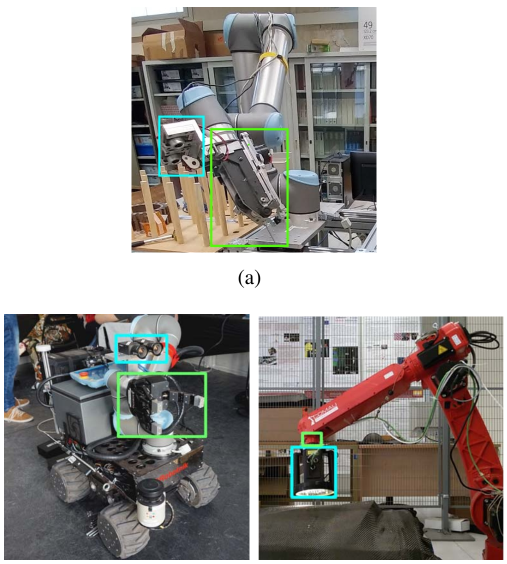


Fig. 1. The proposed two phase procedure. In the first phase, a raw hand-eye calibration is calculated with images grabbed while the robot is moving in the robot base or world reference system. Hence, the checkerboard can be localized. In the second phase, a refined hand-eye calibration is calculated using images grabbed with the robot moving in the checkerboard reference system.



(a)

(b) (c)

Fig. 2. The robot setup exploited in this paper is similar to that of the project EuRoC (a). Other two hand-eye configurations, respectively in the challenge MBZIRC (Desert Lion Team) (b) and the project FibreMap (c), are reported. The hand is in the green box, the eye is in the cyan box.

The remainder of the paper is organized as follows. Section II reviews the work related to the hand-eye calibration problem. Section III describes our novel approach, first giving a picture of the entire workflow, then focusing on the two steps with an eye on the framework adaptability to other robots. In Section IV, our methods are thoroughly evaluated in a simulated and a real environment. Finally, in Section V, conclusions are drawn and future directions of research identified.

II. RELATED WORK

In the past, many methods have been proposed to perform the hand-eye calibration. As reviewed in [1], they can be divided into two categories: approaches coupling hand-eye calibration with conventional robot kinematic model calibration like [14] and approaches decoupling hand-eye calibration from conventional robot kinematic model calibration like [15]. In [1], a new approach belonging to the second category is also proposed, which is faster and more accurate. This is now a common approach implemented in ROS and ViSP.

In [5], Tsai et al. discussed what are the main sources of error. In order to keep the errors low, they suggested to move the robot in different positions, the so-called stations, for each of which the robot is stopped and an image is acquired. They are chosen with a star-drawing technique giving a systematic way of generating an arbitrary number of view points with varying camera distances and angles. In our work, we used their calibration function and ensured the acquisition of images with varying distances and angles, but we did not implement their star-drawing technique. As previously discussed, even if the work by Tsai et al. has been a clear step towards the complete task automation, their procedure requires the calibration pattern position to be known, in turn requiring the human intervention. This is overcome by our two phase approach.

More recently, in [16], [17], two extended hand-eye calibration approaches without the requirement of a calibration pattern have been presented. Indeed, in medical applications, an unsterile calibration pattern cannot be used. In [16], feature tracking and a structure-from-motion approach are exploited. This proved to be useful to simplify the calibration process, which has to be done before every surgical procedure. Nevertheless, this approach is not as accurate as the common one. In [17], the approach is claimed to be accurate. Anyway, it uses the surgery instruments with known CAD models as calibration objects. In our scenario, an object with a known CAD model is less practical than a checkerboard.

III. METHODS

This section focuses on the description of our automatic procedure. For ease of understanding, the main reference systems and transformations are depicted in Figure 3. The symbols B, H, C and P are the initials of Base, Hand, Camera and Pattern, the names of the main reference systems. The symbols BH, HE, EP and BP stay for the respective rototranslations from the Base to the Hand, the Hand to the Eye, the Eye to the Pattern and the Base to the Pattern. Our goal is the estimation of BP and HE (in red).

The automatic procedure can start after moving the robot to a starting position from which the calibration pattern is visible. Our calibration pattern is the asymmetrical circle pattern, currently supported in OpenCV [18], but could be also the classical black-white checkerboard with equally spaced squares. This procedure is characterized by two phases, displayed from left to right in Figure 1. In the first phase, a raw hand-eye transformation (HE) is estimated

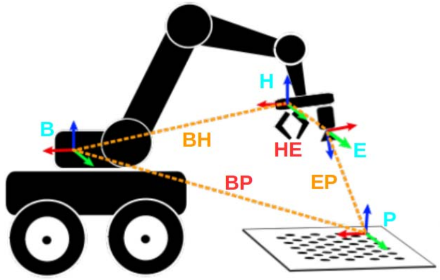
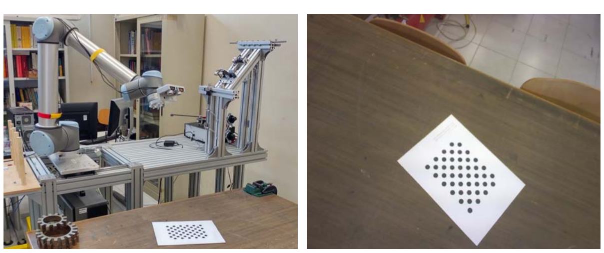


Fig. 3. Representation of the main reference systems (colored triads) and transformations (dot lines). Our goal is the estimation of the Base-Pattern (BP) and the Hand-Eye transformation (HE).

from images grabbed while the arm is moving the hand (H) in the robot base or world reference system (B) with a stop-and-go motion. Indeed, the links BP and HE are both unknown so, instead of moving the camera (E) with respect to the the calibration pattern (P), we can only move the hand (H) with respect to the world/robot base (B). Thus, the actual path of the camera with respect to the checkerboard depends on the starting position. This means that, for instance, the actual number of acquired images framing the calibration pattern and the number of angulations cannot be controlled, impacting the quality of the estimated hand-eye transformation, which, as a result, is also influenced by the starting position. Nevertheless, thanks to this first hand-eye transformation, the checkerboard (P) can be roughly localized. Thus, in the second phase, the robot camera (E) can be moved by the arm with respect to this reference system (P), finally providing a second hand-eye transformation (HE). This second phase gives the advantage of starting the procedure from a known point and moving the arm along a more controlled and repeatable path.

A. Pattern Localization

As previously mentioned, this automatic procedure starts from a position from which the calibration pattern falls in the camera Field of View (FoV). An example is reported in Figure 4. From here, the robotic arm moves through several



(a) (b)

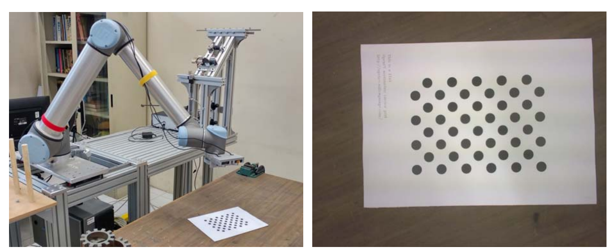
Fig. 4. An example of starting position of the pattern localization: (a) the robotic arm UR10 with a stereo-camera rigidly mounted on the gripper (EuRoC project setup) (b) the checkerboard. The checkerboard is in the camera FoV.

stations with a stop-and-go motion so as to avoid issues when synchronizing the robot-pose with the image. These stations are not chosen a priori. The robotic arm moves its gripper along each axis until the pattern stops being detected. In particular, at fixed steps, a check is performed to verify that the pattern is still detected, hence visible. If so, the acquired image can be used to calibrate and locate the checkerboard, and the robot can keep moving along that axis. Additionally, more stations can be added by rotating the gripper around the gripper axis. Collision avoidance and target reachability is guaranteed by the MoveIt! library.

At the end of this phase, the pattern localization is performed from a set of rectified images $\{I_i\}$ framing the calibration pattern and a set of Base to Hand transformations $\{BH_i\}$ with $i \in \{0, \ldots, N-1\}$. For each image, the transformation Eye to Pattern EP can be estimated from the 3D-2D point correspondences, e.g. by means of the iterative method based on the Levenberg-Marquardt optimization available in the OpenCV library. From $\{BH_i\}$ and $\{EP_i\}$, the hand-eye calibration method by Tsai et al. [1] leads to the estimation of HE. Thus, the pattern calibration location can be easily estimated as $BP = BH_{N-1} \times HE \times EP_{N-1}$, where BH_{N-1} is a valid robot pose and EP_{N-1} the respective valid Eye to Pattern transformation.

B. Automatic Acquisition

Once a first hand-eye HE is estimated and the checkerboard P is located in the work-cell, the camera can be moved with respect to the checkerboard independently from its location in the workspace along a controlled and repeatable path. As shown in Figure 5, the acquisition can always start from the same view of the checkerboard.

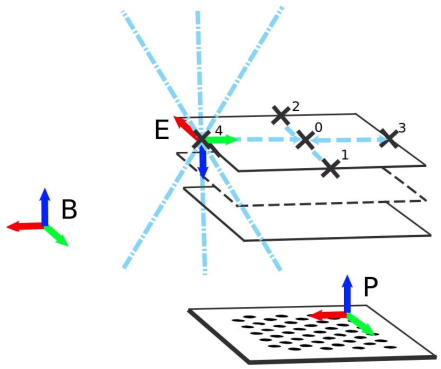


(a) (b)

Fig. 5. An example of starting position of the automatic acquisition: (a) the robotic arm UR10 with a stereo-camera rigidly mounted on the gripper (EuRoC project setup) (b) the checkerboard. The checkerboard is in the camera FoV and clearly visible.

In this phase, the robot moves mimicking a typical acquisition performed by a human and, as suggested by Tsai et al. in [5], varying distances and angles are considered. The robot moves again in a stop-and-go fashion stopping with a fixed stride to check that the calibration pattern keeps being visible. A scheme representing the camera movements is reported in Figure 6. In particular:

- the camera E scans a number of planes *num_p*, which are parallel to the checkerboard but outdistanced of a fixed step *d_p* starting from a minimum distance *d_min_p*. For instance, *num_p* can be equal to 3, *d_p* to $0.05\,\mathrm{m}$ and *d_min_p* to $0.40\,\mathrm{m}$;



(a)

Fig. 6. The camera can be moved with respect to the checkerboard independently from its location in the workspace along a controlled and repeatable path.

- the plane scan is performed from a position 0, almost in the center. Then, the camera is moved along the checkerboard axis x and y, between the position 1 and 2 or 3 and 4. The terminal positions are not fixed. Indeed, as in the previous phase, the camera is moved along an axis until the checkerboard stops being in the FoV;
- each plane can be scanned with varying configurable angles. This is shown in Figure 6, in which the camera E at the position 4 is rotated with three different angles around the axis y. Rotations around other axes can be set too. For instance, one plane can be scanned several times, first with all the angles set to 0, then setting a rotation angle of $10°$ around the axis x, then $10°$ around the axis y and so on.

Again, collision avoidance and target reachability is guaranteed by the MoveIt! library. If any of the stations cannot be reached, the robot continues moving towards the subsequent one. From the new sets $\{BH_i\}$ and $\{EP_i\}$, the hand-eye calibration method by Tsai et al. [1] leads to the estimation of HE.

C. System Adaptability

This approach has been implemented by means of the libraries ViSP, ROS, MoveIt!, OpenCV and PCL, which are widely adopted by the robotic community. Currently, the MoveIt! library supports 65 robots. Thus, even if this implementation has been tested on the Universal Robot UR10 only, it can be potentially used with many others. We selected an approach based on well-known libraries and frameworks for being able to work with different robots and cameras with almost no need for code and system modifications. In fact, our approach obtains the hand-eye calibration starting solely by camera and robot. Standard procedures integrated in the ROS framework provide robot and camera information to our algorithm. A configuration file stores parameters related to different characteristics of the system.

Camera intrinsic parameters are collected by using a *topic*[6]. In the majority of the sensors available in ROS, the *topic* is published directly by the camera driver, and it can be updated by using a camera calibration procedure already available in ROS. Anyway, if no information about intrinsic parameters is exposed by the camera driver, the message firm is publicly available on the ROS website.

Similarly, the system looks for a robot model to learn its physical structure, know about kinematic chains, and understand dynamics. The physical structure is represented by means of the Unified Robot Description Format (URDF)[7]. In the URDF, the robot is composed of joints and links: part dimensions, motion limits, weights, visual and collision shapes complete the robot description. Kinematic chains are defined inside the Semantic Robot Description Format (SRDF)[8]. Moreover, SRDF is usually accompanied by a series of files providing information about collision avoidance, inverse kinematics, robot controllers. MoveIT! is able to exploit such information simply by referring to a specific kinematic group. In our system, it is possible to select the correct kinematic group by changing a parameter in the configuration file. Dynamics is fundamental for a correct simulation in the Gazebo environment. This information can be integrated directly in the URDF file and it is the basis for obtaining realistic tests with physics engines.

IV. EXPERIMENTAL RESULTS

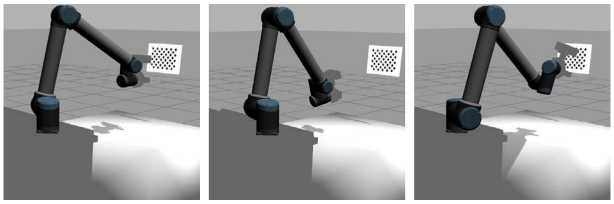
To test the automatic procedure, a scenario similar to the one of the EuRoC project has been selected. The robotic arm is the Universal Robot UR10, collaborative and with 6 degrees of freedom. A stereo-camera composed of two PC webcams is rigidly mounted on it. One of them is taken as a reference, and the hand-eye calibrations are computed with respect to it. In the following, the results of experiments performed in both simulated and real test-beds are discussed.

A. Simulation Experiments

Thanks to the simulation environment, the ground-truth hand-eye transformation and checkerboard can be easily retrieved. To assess the generality of the system, it was tested with the robot starting from 3 different positions (Test A) and with 3 different checkerboard positions in the work-cell (Test B). The 6 different configurations are reported in Figure 7 and 8, respectively. In total, 12 hand-eye transformations and 6 checkerboard locations have been estimated.

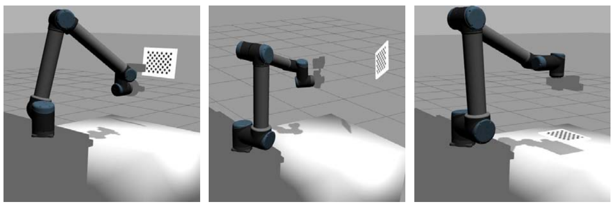
With regard to Test A, the main results are summed up in Table I. The hand-eye transformations have been compared in both translation and angle, in particular Δt is the Euclidean distance in meters while Δq the difference vector between two quaternions in degrees. In addition to the ground truth, the HE transformations have been compared with a baseline consisting in the HE calculated with a manually determined motion defined by 10 waypoints at varying heights and angulations. The baseline, the hand-eye calculated in the first

[6] `http://wiki.ros.org/Topics`
[7] `http://wiki.ros.org/urdf`
[8] `http://wiki.ros.org/srdf`



(a) (b) (c)

Fig. 7. Different starting positions with varying inclinations of the camera. The calibration pattern stays on a vertical plane.



(a) (b) (c)

Fig. 8. Different checkerboard positions: on a vertical, horizontal and inclined plane.

phase and the hand-eye calculated in the second phase are almost equivalent with a standard deviation of only $0.0017\,\text{m}$ and $0.1016°$. Interestingly, in this ideal simulated scenario, the second hand-eye calibration leads to worse performance even if in the second phase the camera is closer to the calibration pattern. This implies that the image quality is higher but also that the number of acquired images is minor since the camera is staying closer to the checkerboard and moving with a fixed step from one position to the next until the checkerboard is visible. In this ideal scenario, the second effect is clearly prominent. Finally, as shown in Table II, the differences in the checkerboard localization are also minimal. The worst entry deviates from the ground-truth of about $0.0159\,\text{m}$ (Position 2) and $1.1412°$ (Position 1). As expected, these deviations are higher since the error on the hand-eye is accumulated with the errors in the robot positioning (BH_{N-1} transformation) and, mainly, the error in the checkerboard positioning with respect to the camera (EP_{N-1} transformation).

With regard to Test B, the main results are summed up in Table III. Even if the robot starts from 3 different positions, both phases of the calibration procedure succeed giving results similar to those already presented.

B. Real Experiments

The entire procedure has been executed from 5 different positions, which have been randomly chosen around the checkerboard at a distance of about 1.00 m. In the second phase, the robot has scanned 3 planes at the distances of 0.40 m, 0.45 m and 0.50 m, from which the checkerboard is better focused. One of these trials has been already shown in Figure 4 and 5. In total, 10 hand-eye transformations, 5 for each of the two phases, and 5 checkerboards have been estimated. In Table IV, the main results are recapped. Given the lack of the ground-truth, no direct comparison can be performed. Anyway, it can be seen that, as expected, the standard deviations of the hand-eye translations and rotations obtained after the first phase (respectively $0.0113\,\text{m}$ and $1.8703°$) are much higher than the standard deviations in the second phase (respectively $0.0054\,\text{m}$ and $0.5324°$). This is due to the fact that the second phase moves the robot along a known path in the checkerboard reference system.

V. CONCLUSIONS

This paper presented an automatic system for localizing the calibration pattern and performing the hand-eye calibration taking a step forward in comparison to existing methods. Indeed, previous works focused on optimizing the hand-eye calibration instead of trying to fully automatize the procedure from the beginning to the end. The proposed approach is based on two phases, in each of which an hand-eye calibration is performed. The second phase allows acquiring images in a controlled setup by moving the camera with respect to the calibration pattern. Both phases were tested in simulated and real scenarios showing that stable results can be obtained after the second phase. As a final contribution to the research community, the toolbox is released online. We plan to use this tool in the upcoming projects and keep on improving it. We would like to test and compare different tool and camera paths in the two phases in order to investigate on the relation between path and overall accuracy.

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TABLE I

HAND-EYE TRANSFORMATIONS IN THE SIMULATION EXPERIMENT TEST A: TRANSLATIONS AND ROTATIONS IN QUATERNIONS. THE ROBOT STARTS FROM THREE DIFFERENT POSITIONS. THE CHECKERBOARD IS VERTICAL.

HE	x	y	z	qx	qy	qz	qw	Δt	Δq
Ground truth	0.0842	0.0959	0.1139	0.5879	-0.3928	0.5879	-0.3928	0.0000	0.0000
Baseline	0.0887	0.0936	0.1087	0.586417	-0.390973	0.589555	-0.39445	0.0072	1.5200
Phase 1 - Position 1	0.0879	0.0914	0.1133	0.5880	-0.3928	0.5878	-0.3929	0.0058	1.4806
Phase 1 - Position 2	0.0866	0.0940	0.1105	0.5888	-0.3929	0.5886	-0.3904	0.0045	1.5720
Phase 1 - Position 3	0.0884	0.0916	0.1119	0.5877	-0.3943	0.5868	-0.3935	0.0063	1.2554
Phase 2 - Position 1	0.0927	0.0945	0.1128	0.5885	-0.3931	0.5875	-0.3923	0.0086	1.4433
Phase 2 - Position 2	0.0924	0.0942	0.1130	0.5886	-0.3930	0.5876	-0.3922	0.0084	1.4893
Phase 2 - Position 3	0.0931	0.0942	0.1126	0.5885	-0.3931	0.5875	-0.3923	0.0091	1.5119

TABLE II

BASE-CHECKERBOARD TRANSFORMATIONS IN THE SIMULATION EXPERIMENT TEST A: TRANSLATIONS AND ROTATIONS IN QUATERNIONS. THE ROBOT STARTS FROM THREE DIFFERENT POSITIONS. THE CHECKERBOARD IS VERTICAL.

BP	x	y	z	qx	qy	qz	qw	Δt	Δq
Ground truth	0.8000	1.5000	1.4000	-0.7071	0.0000	0.0000	0.7071	0.0000	0.0000
Position 1	0.7997	1.4965	1.3977	-0.7066	0.0002	-0.0002	0.7075	0.0042	1.1412
Position 2	0.7864	1.4928	1.3957	-0.7082	-0.0011	-0.0015	0.7059	0.0159	1.1248
Position 3	0.8012	1.4929	1.3964	-0.7063	0.0011	0.0016	0.7078	0.0080	1.1411

TABLE III

HAND-EYE TRANSFORMATIONS IN THE SIMULATION EXPERIMENT TEST B: TRANSLATIONS AND ROTATIONS IN QUATERNIONS. THE CHECKERBOARD IS POSITIONED IN 3 DIFFERENT PLACES: VERTICAL, HORIZONTAL AND ON AN INCLINED PLANE.

HE	x	y	z	qx	qy	qz	qw	Δt	Δq
Ground truth	0.0842	0.0959	0.1139	0.5879	-0.3928	0.5879	-0.3928	0.0000	0.0000
checkerboard 1	0.0927	0.0945	0.1128	0.5885	-0.3931	0.5875	-0.3923	0.0086	1.4433
checkerboard 2	0.0928	0.0943	0.1129	0.5885	-0.3930	0.5875	-0.3922	0.0088	0.9978
checkerboard 3	0.0914	0.0943	0.1130	0.5885	-0.3932	0.5875	-0.3924	0.0074	1.3638

TABLE IV

HAND-EYE TRANSFORMATIONS IN THE REAL EXPERIMENTS AND THEIR STANDARD DEVIATIONS IN THE TWO PHASES.

HE	x	y	z	qx	qy	qz	qw	std(t)	std(q)
Phase 1 - Position 1	-0.2394	-0.0280	-0.0961	0.5141	0.4865	0.5205	0.4776		
Phase 1 - Position 2	-0.2527	-0.0341	-0.0880	0.4856	0.5081	0.5162	0.4894		
Phase 1 - Position 3	-0.2394	-0.0237	-0.1095	0.5067	0.4917	0.5152	0.4859	0.0113	1.8703
Phase 1 - Position 4	-0.2452	-0.0293	-0.0934	0.4993	0.5001	0.5183	0.4816		
Phase 1 - Position 5	-0.2372	-0.0368	-0.0965	0.5004	-0.4933	-0.5223	-0.4831		
Phase 2 - Position 1	-0.2521	-0.0215	-0.1075	0.4974	0.5055	0.4783	0.5180		
Phase 2 - Position 2	-0.2500	-0.0191	-0.1091	0.4973	0.5031	0.4775	0.5211		
Phase 2 - Position 3	-0.2450	-0.0289	-0.1117	0.4950	0.5093	0.4807	0.5143	0.0054	0.5324
Phase 2 - Position 4	-0.2517	-0.0181	-0.1096	0.4960	0.5059	0.4745	0.5225		
Phase 2 - Position 5	-0.2505	-0.0194	-0.1096	0.4974	0.5054	0.4763	0.5199		

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